

Feedback on COMP40018A Exam

Question 1

Exam summary to follow.

Question 1 - Average performance: B

This year's exam question on Discrete Maths has been of average difficulty with respect to previous years, but had certain parts that a lot of students found difficult.

- a) This part asked for some set construction, testing the understanding of the basic set operations; most students got this right.
- b) This was a variant of an exercise in the notes; most students got this right. Although the questions explicitly stated "You do not need to give a formal proof", many students did, and wrote down answers that were too long. Most lost marks for not applying De Morgan's laws correctly, and skipping steps in the proof.
- c) This part was a variant of a question that appeared in the coursework, and was the lowest scoring part of the exam, mainly because many did not see that V_1 is the emptyset, and V_0 is the set of all primes, so largest. Most also did not realise that $\sqsubseteq$ is a relation on numbers, where 1 is minimal, and 0 is maximal.
- d) This was an exercise from the notes, and most students remembered the structure of the proof.
- e) This part was a small variant of coursework, and could be answered through results shown in the notes; most who attempted it had difficulty writing a complete answer, for example by attempting a listing diagram that was not well defined nor correct.

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Question 1

This question's primary remit was to assess the ability to work with propositional and first-order logic. A secondary remit was the precise development of the proofs involved in these areas. The questions were relatively standard, and the class did very well.

Question 1 - Average performance: A*

Here are some more details:

1a)

i) Everyone did pretty well in this part. Some removed too many brackets for sub-part (A) forgetting that the $\leftrightarrow$ has stronger binding than $\neg$ and hence the brackets around $(q \leftrightarrow s)$ could be dropped. For sub-part (B) only the outer brackets could be removed which is what most answered correctly.

ii) Most students answered this correctly. Some common mistakes included using equivalences instead of a direct argument. Referring to bottom and top as truth values; these are connectives! Some missed providing justifications to the inferences being made to why a subformula was true or false. Students should refer to the semantics of the operators. There were some solutions that considered every situation. This was accepted although lengthy and unnecessary and should be avoided.

iii) Overall answers followed the correct structure. Some common mistakes included not referencing an equivalence law that was given, applying several different rules in a single step, or giving a final formula not in CNF.

iv) Students overall recognised that the system was not sound but struggled to give a correct justification. The justification here requires showing a single counterexample. This counterexample is a situation in which you have $A \rightarrow B \vee C$ true and A true but B false.

1b)

i) This question required students to demonstrate an understanding of FOL syntax. Common errors included splitting the quantifier and variable (e.g. $\forall x$) into two separate nodes, creating leaf nodes for variables appearing as arguments in a predicate.

ii) This question required students to demonstrate an understanding of FOL semantics and determining when formulas are true in an L-structure. Students did exceptionally well in this question. Some missed giving a justification as to why a formula was not true and this should be in the form of assignments to the variables.

iii) This question tested the correct application of the natural deduction inference rules. Common mistakes included forgetting to add an assumption box for the forall const, misapplying the or-elimination rule, or misapplying the forall introduction rule.

Question 2

This question's primary remit was to assess the ability to prove a property by induction, write program specifications and prove correctness of imperative programs. A secondary remit was the precise development of such proofs. This question was not too challenging, but there were some tricky details in the program specifications. Overall, the class did very well, although a few students clearly struggled to budget their time effectively between the two questions on this exam paper.

Question 2 - Average performance: B

Here are some more details:

2a) This part of the question involved proving a property about Tetrahedral numbers by Mathematical Induction. This proof wasn't too challenging, so the assessment focussed on the ability to stick to a clear proof structure, and to ensure that all steps of the proof were well justified. The majority of students did this well. The three most common mistakes were:

- forgetting to state what is being shown in the Base Case;
- forgetting to remove the $\forall n \in N$ quantification outside the $T(n)$ property in the Inductive Hypothesis (so you end up assuming the property we are trying to prove overall);
- skipping the term collection, or expanding all of the brackets in the proof of the Inductive Step (thus making the proof very hard to follow).

2b) This part of the question was a simple application of the property proved in the previous part. Almost all students got this right, with a few silly calculation errors being the cause of most mistakes. Thankfully, no-one tried to compute $T(100)$ recursively!

2c) This part of the question involved writing mid-conditions, loop invariants and loop variants for the nested loops in the **tetra** function. This was something a bit different to what was seen in the lectures, but the key was to approach each loop separately and think carefully about the provided $T(n)$ function. The outer loop was moving from term to term of $T(n)$, while the inner loop was counting up

the sum of $\sum_{k=1}^{i-1} k$.

Most students answered this very well (scoring over half of the marks), but there were a few common mistakes in specifying the upper bounds of **cur** (which was $x + 1$ in the *outer* loop, but x in the *inner* loop) and **i** (which was **cur** + 1). Similarly, **tot** stored the value $T(\mathbf{cur} - 1)$ at the beginning/end of the *outer* loop, but $T(\mathbf{cur})$ at the mid-condition **M**, after the *inner* loop.

A lot of students also forgot to include a bound on **cur** in the loop invariant I_2 for the *inner* loop, or tried to include **i** in the loop invariant I_1 for the *outer* loop (where no such variable is in-scope).

The loop variants, by contrast, were nearly always stated correctly (I was generous in accepting any decreasing term), though some students did try to write Boolean Expressions here, rather than Integer Expressions.

2d) This part of the question involved proving that the *outer* loop in the **tetra** function re-establishes the invariant I_1 . To make this feasible within an exam setting, the question broke the overall proof into two parts and assumed that the inner loop was totally correct.

A lot of students skipped this part, or ran out of time for it, but those that did attempt it were mostly able to set up the proofs correctly (there were a few mistakes with the use of *old* annotations, which were not required at all for the first proof). There were a few cases where students set-up the proofs the wrong way around (trying to show that I_i was established before the *inner* loop, or that M_i holds at the end of the *outer* loop). It always pays to read the question carefully.

Those students that attempted the full proofs tended to perform better on the second proof than the first. I assume that this was due to the second proof being very similar to examples seen in the

lectures. The key step in the first proof was to spot that the sum $\sum_{k=1}^{i-1} k$ is empty (thus equal to 0) when

$i = 1$. The required bounds on **cur** and **i** follow fairly easily from the given loop invariant and loop condition.

Extra

For the interested, I asked Co-Pilot to help me write a new song for Olivia, but it struggled to understand the progression of the Tetrahedral numbers or to keep the song consistent and avoid repetition. Still, given below are a couple of snippets from the song that it helped me to create. Happy Techmas!

On the twenty-fifth day of Techmas, my true love gave to me: Twenty-five virtual reality headsets immersing, twenty-four AR glasses displaying, Twenty-three smart wearables syncing, twenty-two smart garden sprinklers spraying, Twenty-one smart home sensors sensing, twenty AI assistants assisting, Nineteen smart cameras recording, eighteen Wi-Fi routers routing, Seventeen smart thermostats adjusting, sixteen power banks charging, Fifteen USB drives storing, fourteen Bluetooth earbuds syncing, Thirteen wireless chargers charging, twelve smart bulbs glowing, Eleven smart locks locking, ten smart fridges chilling, Nine e-readers reading, eight fitness trackers tracking, Seven drones a-flying, six smart speakers chatting, Five robot vacuums! Four noise-cancelling headphones, Three smartwatches, two VR goggles, And a brand-new laptop, shiny and sleek. That's 325 gifts today, and 2,925 so far, but watch out caus here come some more!

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On the hundredth day of Techmas, my true love gave to me: One hundred AI-generated poems, ninety-nine quantum computers crunching, Ninety-eight biometric scanners scanning, ninety-seven drones a-delivering, Ninety-six AR contact lenses displaying, ninety-five virtual worlds exploring, Ninety-four holographic displays shimmering, ninety-three exoskeletons assisting, Ninety-two neural networks learning, ninety-one nanobots repairing, Ninety smart cities optimizing, eighty-nine blockchain transactions validating, Eighty-eight self-driving cars navigating, eighty-seven chatbots conversing, Eighty-six space elevators ascending, eighty-five bionic limbs enhancing, Eighty-four gene-editing therapies curing, eighty-three fusion reactors powering, Eighty-two quantum teleporters beaming, eighty-one brain-computer interfaces connecting, Eighty smart homes automating, seventy-nine asteroid miners extracting, Seventy-eight solar farms harvesting, seventy-seven Mars colonies thriving, Seventy-six time machines whirring, seventy-five cyborg pets playing, Seventy-four quantum encryption keys securing, seventy-three holographic concerts rocking, Seventy-two AI chefs cooking, seventy-one 3D-printed organs replacing, Seventy smart glasses augmenting, sixty-nine lunar bases pioneering, Sixty-eight nanoscale factories producing, sixty-seven drone swarms surveilling, Sixty-six fusion-powered yachts sailing, sixty-five quantum entanglement experiments enthralling, Sixty-four AI poets rhyming, sixty-three brainwave headbands meditating, Sixty-two quantum AIs debating, sixty-one Mars rovers exploring, Sixty smart farms optimizing, fifty-nine space tourism launches thrilling, Fifty-eight underwater habitats thriving, fifty-seven neural lace implants integrating, Fifty-six asteroid mining guilds collaborating, fifty-five quantum supremacy claims proving, Fifty-four exoplanets discovered, fifty-three biohacker collectives biohacking, Fifty-two lunar resorts enchanting, fifty-one AI artists creating, Fifty smart factories automating,

forty-nine quantum algorithms solving, Forty-eight cybernetic enhancements upgrading, forty-seven fusion-powered jets soaring, Forty-six AR museums enlightening, forty-five neural implants enhancing, Forty-four quantum simulations predicting, forty-three Mars terraforming projects blooming, Forty-two AI therapists counseling, forty-one space elevators ascending, Forty smart classrooms educating, thirty-nine quantum communication networks connecting, Thirty-eight brainwave pianists composing, thirty-seven nanobots healing, Thirty-six VR theme parks thrilling, thirty-five exosuit racers competing, Thirty-four holographic novels captivating, thirty-three AI comedians jesting, Thirty-two quantum cryptographers securing, thirty-one Mars greenhouses growing, Thirty smart hospitals diagnosing, twenty-nine asteroid mining expeditions launching, Twenty-eight neural net dream analysts decoding, twenty-seven quantum chess grandmasters strategizing, Twenty-six AR art galleries mesmerizing, twenty-five bionic athletes sprinting, Twenty-four AI-generated symphonies harmonizing, twenty-three brainwave sleep inducers soothing, Twenty-two quantum stock traders predicting, twenty-one Mars terraforming milestones reaching, Twenty smart laboratories researching, nineteen space telescopes observing, Eighteen neural network fashion designers styling, seventeen quantum teleportation breakthroughs achieving, Sixteen AR fashion influencers posing, fifteen bionic dancers twirling, Fourteen AI-generated love songs serenading, thirteen brainwave meditation apps calming, Twelve quantum weather forecasters predicting, eleven Mars habitats thriving, Ten smart cities optimizing, nine asteroid mining corporations prospering, Eight neural network poets rhyming, seven quantum encryption protocols securing, Six AR travel guides narrating, five bionic hearts beating, Four AI-generated masterpieces painting, three brainwave mindfulness apps centering, Two quantum entangled hearts connecting, And a brand-new laptop, shiny and sleek.

That's 5,050 gifts today, and 171,700 in all, but I don't have room to move any more!